Predictors of sPCI–pPCI Discordance: OLS Regression Analysis

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Source: `notebooks/02-assoc_difference.ipynb`

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1. Background and Objective

A systematic positive bias between intraoperative surgeon PCI (`sPCI`) and preoperative/pathological PCI (`pPCI`) has been established (mean sPCI − pPCI ≈ 6.9, see report `01-agreement_all_report.md`). This analysis investigates **which patient and clinical characteristics explain the magnitude of this discordance**, using ordinary least squares (OLS) regression.

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2. Study Sample

After excluding tumour type 8, the analytic sample comprised **N = 412** patients. Tumour type 1 (n = 122) was set as the reference category.

Predictors included:

Variable	Type	Notes
Sex	Binary	0 = female, 1 = male
Age	Continuous	Years
Zn HIPEC	Binary	Zinc-based HIPEC regimen
PreOP CTx	Binary	Any preoperative chemotherapy (≥1 cycle)
Thermoablation	Binary	Intraoperative thermoablation performed
Tumour type	Nominal	Types 2–7 vs. reference type 1

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3. Outcome Variables

Two outcomes were constructed:

- **`raw_diff`** = sPCI − pPCI (signed discordance)
- **`abs_diff`** = |sPCI − pPCI| (magnitude of discordance)

Statistic	raw_diff	abs_diff
Mean	7.09	7.39
SD	7.00	6.68
Median	5.0	5.0
Min	−8	0
Max	32	32

The positive mean and median of `raw_diff` confirm that surgeons consistently find **more** peritoneal disease intraoperatively than predicted preoperatively (median discordance of 5 PCI points). Both distributions depart significantly from normality (Shapiro-Wilk: $p < 0.001$ for both).

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4. OLS Regression Results

Outcome: `raw_diff` (sPCI − pPCI)

	F-statistic	df	p-value
Overall model	10.52	(11, 400)	5.1×10^{-17}

Variance explained: $R^2 = 0.224$, Adjusted $R^2 = 0.203$

The model is highly significant overall and explains approximately **20% of the variance** in discordance.

Regression Coefficients

Predictor	Coefficient	SE	t	p	95% CI
Intercept	7.23	2.44	2.96	0.003	[2.44, 12.03]
Sex (male)	**−1.41**	0.71	−2.00	**0.047**	[−2.79, −0.02]
Age	−0.02	0.03	−0.69	0.492	[−0.07, 0.03]
Zn HIPEC	2.28	1.65	1.38	0.169	[−0.97, 5.52]
PreOP CTx	−0.85	0.83	−1.02	0.309	[−2.49, 0.79]
Thermoablation	**−2.56**	0.76	−3.38	**0.001**	[−4.05, −1.07]
Tumour type 2	−0.08	1.51	−0.06	0.955	[−3.06, 2.89]
Tumour type 3	1.39	1.34	1.03	0.302	[−1.25, 4.03]
Tumour type 4	1.44	1.02	1.41	0.159	[−0.57, 3.45]
Tumour type 5	**+7.05**	0.91	7.73	**< 0.001**	[5.26, 8.85]
Tumour type 6	−0.78	1.16	−0.67	0.503	[−3.06, 1.50]
Tumour type 7	**+5.24**	1.34	3.92	**< 0.001**	[2.61, 7.88]

Key Findings

- **Tumour type 5** is the strongest predictor: on average, these patients show **+7.1 points** more discordance than type 1 ($p < 0.001$).
- **Tumour type 7** shows **+5.2 points** more discordance than type 1 ($p < 0.001$).
- **Thermoablation** is associated with **2.6 points less** discordance ($p = 0.001$), possibly reflecting more localised and predictable disease in patients selected for this procedure.
- **Male sex** is associated with **1.4 points less** discordance than female sex ($p = 0.047$), a modest but statistically significant effect.
- Age, Zn HIPEC, PreOP CTx, and tumour types 2, 3, 4, and 6 are **not significant** predictors.

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5. Residual Diagnostics

Test	Statistic	p-value	Conclusion
Shapiro-Wilk (residuals)	W = 0.956	< 0.001	Residuals non-normal
Jarque-Bera	—	< 0.001	Confirms non-normality
Breusch-Pagan (LM)	39.88	< 0.001	Heteroskedasticity present
Breusch-Pagan (F)	3.90	< 0.001	Heteroskedasticity present
Skewness	0.83	—	Right-skewed residuals
Kurtosis	3.87	—	Slightly leptokurtic

Both OLS assumptions of **normality** and **homoskedasticity** of residuals are violated. The reported standard errors and p-values should be interpreted with caution; they are likely anti-conservative (i.e., p-values may be overstated).

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6. Limitations and Recommendations

1. **Heteroskedasticity-robust standard errors** (HC3) should be computed to obtain reliable inference. This will not change the point estimates but may widen confidence intervals and inflate some p-values.
2. **Quantile regression** on the median would be more robust given the skewed, heteroskedastic outcome and would not rely on distributional assumptions.
3. The signed outcome (`raw_diff`) assumes direction matters. If only magnitude is of interest, rerunning the model on `abs_diff` should be considered.
4. Tumour types with small cell counts (types 2, 6, 7) warrant cautious interpretation due to limited statistical power.

5. The negative coefficient for **Thermoablation** likely reflects **selection bias** rather than a causal effect — patients undergoing thermoablation may have more resectable, well-demarcated disease that is easier to assess preoperatively.

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7. Summary

The OLS model explains ~20% of variance in sPCI–pPCI discordance. The dominant drivers are **tumour type** (especially types 5 and 7, which carry far more intraoperative surprise than type 1), followed by **thermoablation** (associated with less discordance) and **sex** (males slightly less discordant). Age and preoperative chemotherapy do not independently predict discordance. OLS assumptions are violated and robust inference methods are advised before drawing definitive conclusions.